Quizrr Chapter-Wise Test for JEE Main - 2025

Q.1

Triaminobenzene is a

(A) 2° amine

(B) 3° amine

(C) 1° amine

(D) Quarternary salt

Q.2

The colour of p -amino azobenzene is

(A) Orange

(B) Congo red

(C) Bismark brown

(D) Indigo

Q.3

Pure aniline is a

(A) Colourless solid

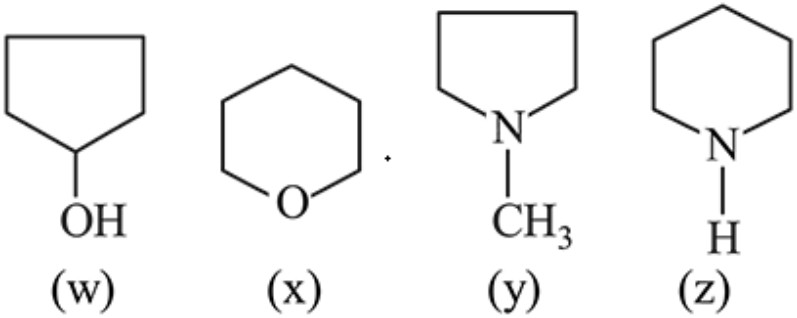
(B) Brown coloured solid

(C) Colourless liquid

(D) Brown coloured liquid

Q.4

Arrange the following $(\mathrm{w}, \mathrm{x}, \mathrm{y}, \mathrm{z})$ in decreasing order of their boiling points:



(w) (x) (y) (z)

(A) $\mathrm{w} > \mathrm{x} > \mathrm{z} > \mathrm{y}$

(B) $\mathrm{w} > \mathrm{x} > \mathrm{y} > \mathrm{z}$

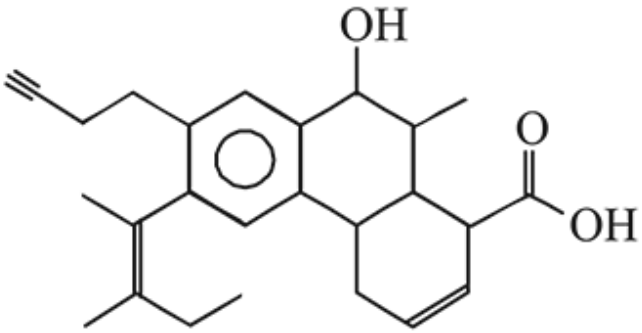
(C) $\mathrm{w} > \mathrm{z} > \mathrm{y} > \mathrm{x}$

@IITJEE_Advanced

(D) $w > z > x > y$

Q.5

How many degrees of unsaturation are there in the following compound?



(A) 6

(B) 7

(C) 10

(D) 11

Q.6

Allyl isocyanide has

(A) 9σ and 4π -bonds

(B) 8σ and 5π -bonds

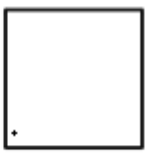
(C) $9\sigma, 3\pi$ and 2 non- bonded electrons

(D) $8\sigma, 3\pi$ and 4 non - bonded electrons

Q.7



Cyclopropane
(I)



Cyclobutane
(II)



Cyclopentane
(III)

The correct order of heats of combustion of above compounds is:

(A) $I > II > III$

(B) $II > I > III$

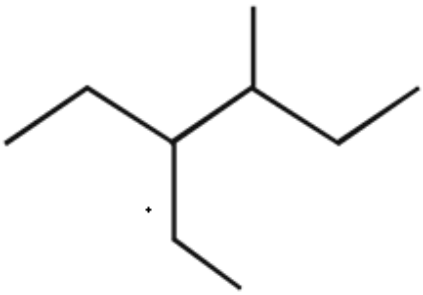
(C) $III > II > I$

(D) $III > I > II$

Q.8

@IITJEE_Advanced

The correct IUPAC name of the following compound



is:

(A) 4 - methyl - 3 - ethylhexane

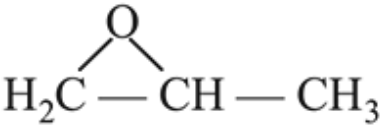
(B) 3 - ethyl - 4 - methylhexane

(C) 3,4 - ethylmethylhexane

(D) 4 - ethyl - 3 - methylhexane

Q.9

The IUPAC name of the compound



is

(A) 1,2 -Propoxide

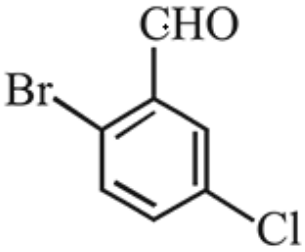
(B) Propylene oxide

(C) 1,2 -Oxo propane

(D) 1,2 -Epoxy propane

Q.10

Which of the following is the IUPAC name of the compound



(A) o— Bromo —m— chlorobenzaldehyde

(B) 2— Bromo —5— chlorobenzaldehyde

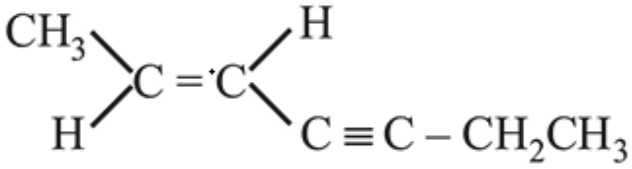
(C) 6— Bromo —3— chlorobenzaldehyde

(D) 1— Bromo —4— chlorobenzaldehyde

@IITJEE_Advanced

Q.11

The IUPAC name of the following compound is



(A) (E)-2-hepten-4-yne

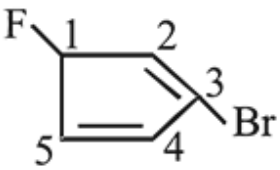
(B) (Z)-5-hepten-3-yne

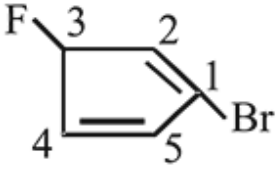
(C) (E)-5-hepten-3-yne

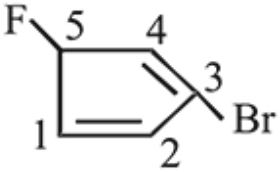
(D) (Z)-2-hepten-4-yne

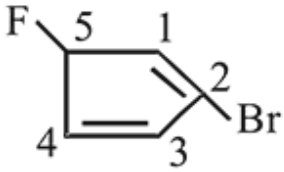
Q.12

Which of the following numberings is correct?

A 

B 

C 

D 

(A) A

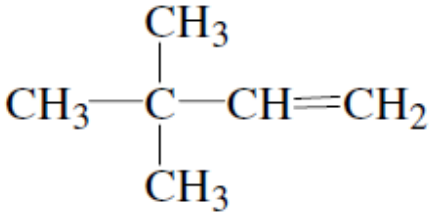
(B) B

(C) C

(D) D

Q.13

The IUPAC name of the compound



is

(A) 3, 3, 3-Trimethylpropane

(B) 1, 1, 1-Trimethylprop-2-ene

(C) 3, 3-Dimethylbut-1-ene

(D) 2, 2-Dimethylbut-3-ene

@IITJEE_Advanced

Q.14

Which one of the following contain isopropyl group?

(A) $2-$Methylpentane

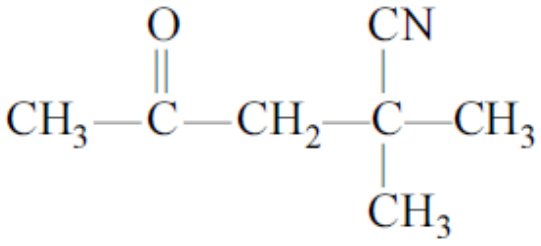
(B) $2,\ 2,\ 3,\ 3-$Tetramethylpentane

(C) $2,\ 2,\ 3-$Trimethylpentane

(D) $3,\ 3-$Dimethylpentane

Q.15

The IUPAC name of the compound



is

(A) 4 -cyano-4-methyl-2-oxopentane

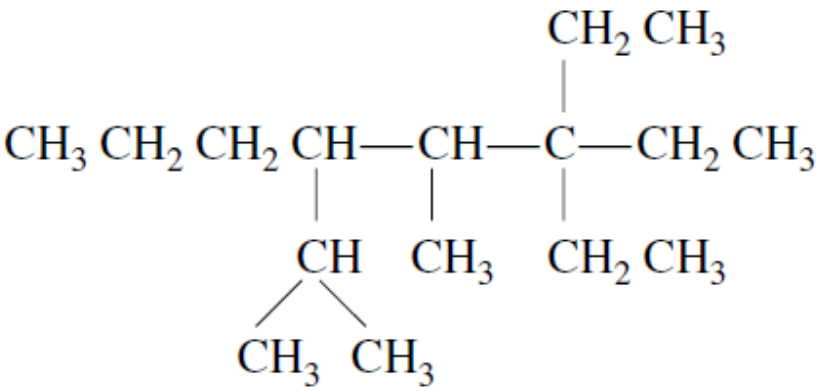
(B) 2 -cyano-2-methyl-4-oxopentane

(C) 2,2 -dimethyl-4 oxopentanenitrile

(D) 4 -cyano-4-methyl-2-pentanone

Q.16

The IUPAC name of the compound



is

(A) 3,3,-diethyl-4-methyl-5-isopropyloctane

(B) 3,3-diethyl-5-isopropyl-4-methyloctane

(C) 4-isopropyl-5-methyl-6,6-diethyloctane

(D) 6,6-diethyl-4-isopropyl-5-methyloctane

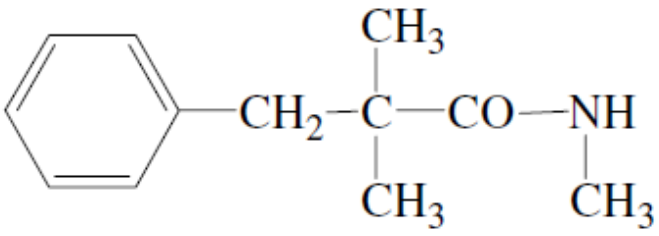
@IITJEE_Advanced

Q.17

Which of the following statements is not correct?

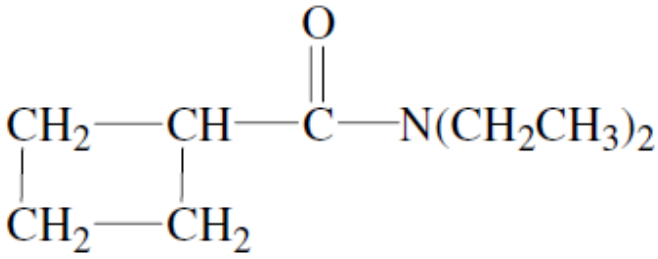
(A) The IUPAC name of $CH_3CH = CHCH = CHCOOH$ is 2,4 -hexadienoic acid.

(B) The IUPAC name of the compound



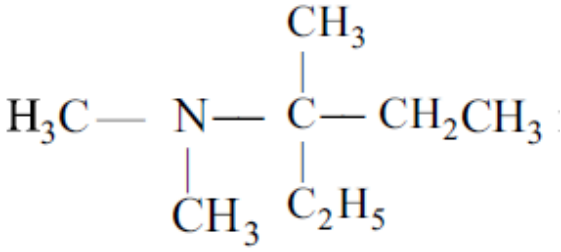
is N-methyl-2,2-dimethyl-3-phenylpropanamide.

(C) The IUPAC name of the compound



is $N - N$-diethylcyclobutanecarboxamide.

(D) The IUPAC name of the compound



is $(N, N$-dimethylamino)-3-ethylbutane.

Q.18

How many of the following compounds can be categorized as allylic halides? 3-Chloropropene, 3 -bromocyclohexene, benzyl chloride, 4 -iodopent- 2 -ene, bromobenzene, (1-chloroethyl)benzene, 4 -chlorocyclopentene, 1-chlorobut-1-ene, 1-chlorobut-2-ene

(A) 1

(B) 3

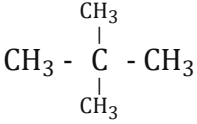
(C) 4

(D) 5

Q.19

What is the name of the following compound ?

@IITJEE_Advanced



(A) Pentane

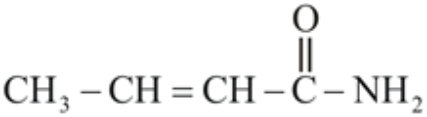
(B) Isobutane

(C) Neopentane

(D) None of these

Q.20

Common name of



, is :-

(A) Butyramide

(B) Acrylamide

(C) Crotonamide

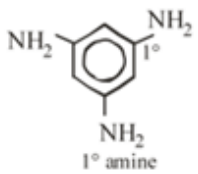
(D) Allylamide

Answers & Solutions

Q.1 Answer:

1° amine

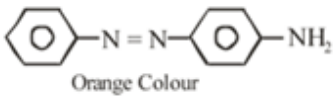
Solution:



Q.2 Answer:

Orange

Solution:



Q.3 Answer:

Colourless liquid

Solution:

Pure aniline is colorless liquid

Q.4 Answer:

@IITJEE_Advanced

$w > z > x > y$

Solution:

B.P. $\propto$ extent of H bonding $\propto$ surface area of molecule.

Q.5 Answer:

11

Solution:

Unsaturation means the presence of double bond or triple bond. It is used to calculate the number of rings and pi bonds present in a molecule.

Degree of unsaturation = 11

Q.6 Answer:

$9\sigma, 3\pi$ and 2 non- bonded electrons

Solution:

Allyl isocyanide has the structure $H_2C = CH - CH_2 - N \vec{C}$

it has 9σ and 3π bonds along with 2 nonbonded electrons

Q.7 Answer:

$III > II > I$

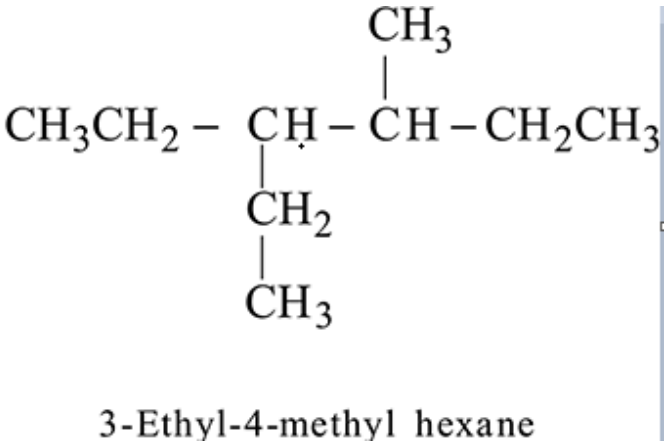
Solution:

More the no. of carbons more will be the heat of combustion.

Q.8 Answer:

3 - ethyl - 4 - methylhexane

Solution:



3-Ethyl-4-methyl hexane

Q.9 Answer:

1,2 -Epoxy propane

Solution:

1,2 -Epoxy propane is the correct IUPAC name of given compound.

Q.10 Answer:

@IITJEE_Advanced

2— Bromo —5— chlorobenzaldehyde

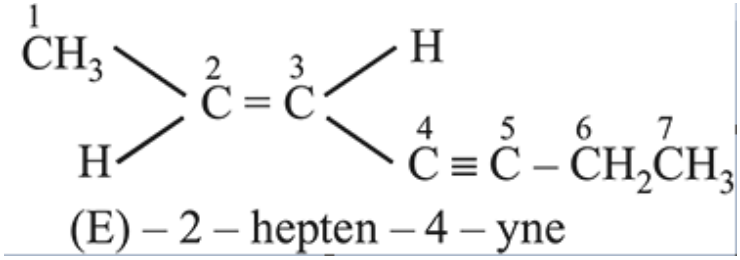
Solution:

The compound is a derivative of benzaldehyde. Start numbering C-atoms of benzene nucleus from C-atom bearing CHO group.

Q.11 Answer:

(E)-2-hepten-4-yne

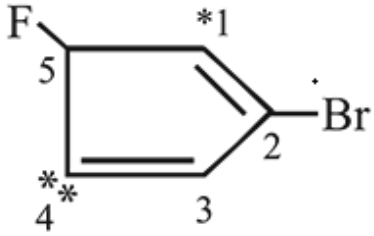
Solution:



Q.12 Answer:

D

Solution:

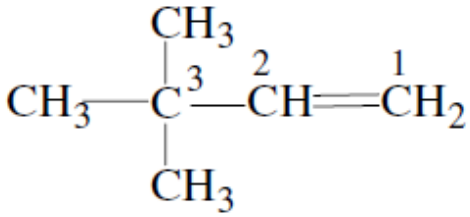


The highest priority functional group is the double bond, and the carbon chain must be numbered such that this functional group is given the lowest number carbon. This is then followed by the lowest numbering of both halogen groups. Therefore, (D) is the correct numbering.

Q.13 Answer:

3, 3-Dimethylbut-1-ene

Solution:



; 3, 3-Dimethylbut-1-ene

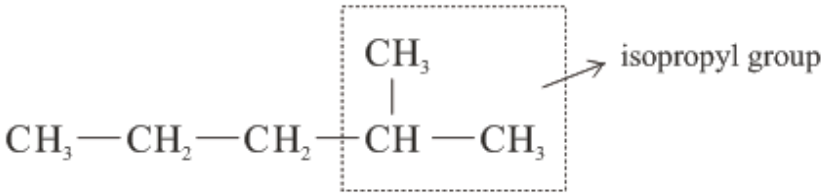
Q.14 Answer:

2 - Methylpentane

Solution:

@IITJEE_Advanced

Isopropyl group $-CHCH_3{}_2$ **is attached to the propyl group in** 2 - methylpentane.



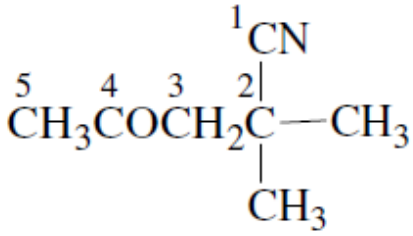
2 - **methylpentane**

Q.15 Answer:

2,2 -dimethyl-4 oxopentanenitrile

Solution:

The numbering goes as follows.



Q.16 Answer:

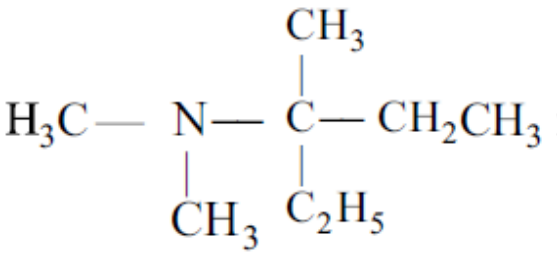
3,3-diethyl-5-isopropyl-4-methyloctane

Solution:

3,3-diethyl-5-isopropyl-4-methyloctane

Q.17 Answer:

The IUPAC name of the compound



is $(N, N$-dimethylamino)-3-ethylbutane.

Solution:

The correct name is $3-(N, N$**-dimethylamino)-3-methylpentane**

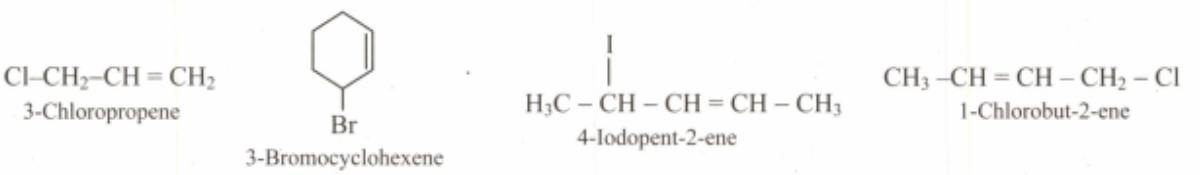
Q.18 Answer:

4

Solution:

In allylic halides, halogen atom is bonded to a sp^3 **hybridised** C**-atom next to carbon carbon double bond.**

@IITJEE_Advanced



Q.19 Answer:

Neopentane

Solution:

Neopentane $CH_3 - \underset{\underset{CH_3}{|}}{\overset{\overset{CH_3}{|}}{C}} - CH_3$

Branched chain compound.

Q.20 Answer:

Crotonamide

Solution:

$CH_3 - CH = CH - \underset{\text{functional group}}{\overset{*}{\underset{\uparrow}{C}}}\ \text{carbon} \rightarrow known\ \ as\ \ croton \therefore CH_3 - CH = CH - \overset{O}{\overset{\|}{C}} - NH_2\ \ is\ \ named\ \ as\ \ crotonamide.$

@IITJEE_Advanced